

AANCHAL MAURYA

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RESEARCH INTEREST

I am broadly interested in Computer Vision, Deep Learning, and Generative Models, with a focus on applying learning-based methods to visual understanding, image processing, and real-world perception tasks.

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT) <i>B.Tech, Electronics and Communication Engineering</i> CGPA: 7.98/10 Parth Institute <i>JEE Mains – 98.60 Percentile</i> Delhi Public School, Panipat Refinery <i>Class XII – 94.0%</i>	Surat, India 2024 – Present Vadodara, India 2023 – 2024 Panipat, India 2024
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EXPERIENCE

Summer Research Intern <i>Indian Institute of Technology (IIT) Mandi Visual Intelligence and Machine Learning (VIML) Lab</i> • Worked on semi-supervised video anomaly detection by adapting a research pipeline to the NWPU-Campus dataset using PyTorch. • Prepared and integrated the dataset by implementing preprocessing, frame extraction, object tracking, model training, and evaluation workflows. • Reproduced experimental results from a research paper while debugging and optimizing the training pipeline on Linux with CUDA, PyTorch, OpenCV, and MMCV.	Mandi, HP, India May 2026 – Present
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ACHIEVEMENTS

NTIRE 2026 Controllable Aperture Bokeh Rendering Challenge (CVPR Workshop) – Participated as part of a team; results awaited.
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TECHNICAL SKILLS

Programming Languages: Python, C, C++ Deep Learning & CV Frameworks: PyTorch, TensorFlow, Keras, OpenCV Libraries & Tools: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn Deployment & Dev Tools: Docker, Git/GitHub, Hugging Face Spaces, Google Colab, Jupyter Notebook
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PROJECTS

Neural Style Transfer <i>PyTorch, VGG19, CNNs</i> GitHub Demo • Implemented neural style transfer using a pre-trained VGG19 network, jointly optimizing content and style loss functions across multiple feature layers. • Encoded style through Gram matrix computation over multi-level feature maps, enabling texture transfer independent of content structure. • Tuned loss-weight hyperparameters and learning rate schedules, improving visual coherence and convergence stability across diverse image pairs.
Emotion Detection System <i>CNNs, PyTorch / TensorFlow, OpenCV</i> GitHub Demo • Designed and trained a CNN-based facial emotion recognition system on grayscale datasets, achieving multi-class classification across 7 emotion categories. • Built a preprocessing pipeline with face detection, normalization, and augmentation to improve generalization under varied lighting and pose conditions. • Deployed a lightweight real-time inference demo with live webcam-based emotion prediction.
Weather Anomaly Detection <i>PyTorch, ResNet, Statistical Modeling</i> GitHub • Built a deep learning pipeline to detect extreme meteorological events, framing the task as a supervised anomaly classification problem on simulated weather data. • Modeled realistic sensor noise using Poisson and Gamma distributions; incorporated ResNet-style skip connections, yielding a 14% recall improvement and 9% F1-score increase over baseline.

POSITIONS OF RESPONSIBILITY

Co-Head <i>MindBend TechFest</i>	SVNIT, Surat
Executive <i>Google Developer Groups on Campus (GDGC), NIT Surat</i>	SVNIT, Surat